

Kevin Corbett

Project 3 Milestone2

Introduction

Had to do a lot of refactoring, I knew something must have been off in my M1 submission because it took ~20 epochs to get similar accuracy to others. A few issues I had to fix: incorrect transpose matrix multiplication, a few incorrect dimensions in my matrix multiplications (totally ignoring the image length for one of the weights), and not accumulating deltas during batches. Pretty remarkable that with enough epochs it was able to get to >90% accuracy.

Performance

Version	Tile Size	Core Count	256 Time(s)	512 Time(s)	256 Grind Rate	512 Grind Rate	Accuracy 256	Accuracy 512
CPU	64	1	104.35	105.42	2,885	2,857	.91	.88
OpenMP No BLAS	64	16	28.02	17.08	10,714	17,647	.91	.88
BLAS	64	1	26.91	26.0	11,111	11,538	.91	.88
GPU	N/A	N/A	5.15	4.06	58,252	73,892	.91	.88